

Questions on RESfM

For Fadi Khatib — re: U-ESFM (unsupervised / self-supervised outlier handling for equivariant SfM). Ordered by how much each answer would change what I run next. Questions 1–5 are the ones I would most value; the rest are useful but not blocking.

A. Reproducing your results and your baselines

1. Which versions and settings did you use for GLOMAP, Theia and COLMAP, and how were our precomputed point tracks passed into them?

Section 4.3 says all methods ran on the same tracks, but these pipelines normally build their own tracks from features and matches, so the ingestion path is a real interface decision. GLOMAP is the baseline I most need to reproduce faithfully, since on 1DSfM it beats RESfM on rotation (1.27° vs 3.98°) while losing on translation. If my reproduction diverges I would like to know whether to suspect my setup or the settings.

2. Running your released checkpoint on a few test scenes, roughly what gap to the published numbers would you consider normal rather than a sign that my evaluation path is wrong?

I am treating a reproduction gate as blocking before anything else. Knowing your expected tolerance would stop me chasing a difference that is just environment.

3. Were the classical baselines run with their own internal outlier filtering enabled, or disabled so that the comparison isolates the track contamination?

B. Evaluation conventions

4. You told me the priority is translation error, then rotation error, then number of registered cameras. Is that lexicographic or weighted?

That is, does translation decide the winner outright with the others as tie-breaks, or can a large rotation loss outweigh a small translation win? This directly determines how I state my main claim, since my current cross-dataset result is strongest on rotation.

5. When methods register different numbers of cameras, do you compare errors over each method's own registered set, or over the intersection?

On 1DSfM RESfM registers 72% of cameras and GLOMAP 98%, so the two error figures are not over the same cameras. I want to follow whatever convention you consider correct rather than inventing one.

6. Tables 2 and 10 have no Mean row for 1DSfM. Is there a reason you did not aggregate that dataset, or is per-scene simply the convention there?

I have been averaging your ten scenes myself and want to know whether to present it that way or stay per-scene.

7. Section 4.1 says 14 test scenes were selected from MegaDepth Group 1, but Table 1 has 13 rows above the middle rule. Was one scene dropped?

C. Training data and label generation

8. What was the outlier percentage distribution across the 27 training scenes and 4 validation scenes?

The paper reports contamination for every test scene but not for training. I need this to say anything defensible about whether the outlier head was exposed to high-contamination regimes, versus the alternative explanation that its labels degrade there.

9. How reliable do you consider the COLMAP-derived labels on the most contaminated scenes?

The Appendix C procedure depends on COLMAP's reconstruction, and per-observation labelling is less noise-tolerant than pose estimation. Did you see label quality vary with contamination, and did that influence the

choice of the 4-pixel threshold?

10. Did you ever try regenerating labels on a target domain such as 1DSfM and retraining the outlier head there?

This is the obvious steelman baseline against any claim that the supervised head does not transfer, and I would rather learn from you whether it has been tried than spend a week on it.

D. Design decisions I may need to discuss in the paper

11. Appendix B sets the removal threshold to 0.8 for Strecha and BlendedMVS instead of 0.6. How was that chosen, and how sensitive are the results to it?

I am considering an argument that per-domain threshold selection assumes prior knowledge of the contamination level. I would rather understand your reasoning first than characterise it from the appendix alone.

12. How much of the outlier robustness comes from the learned head versus the robust BA?

Table 11 shows robust BA is doing a great deal on its own (Tower of London: 0.67° vs 57.41°). Did you ablate a version with the head replaced by a purely statistical rule such as a MAD or percentile threshold on reprojection error? If a statistical rule matches the learned head, that reframes my paper considerably.

13. Do you have a sense of the ceiling — how the method performs with oracle inlier/outlier labels at inference?

This would tell me whether the outlier mechanism is the bottleneck at all, or whether both supervised and self-supervised variants are already near the achievable limit.

14. On the excess-removal limitation in your conclusion: is a case like Tower of London (94 of 467 cameras) a known failure mode with a diagnosis, or still open?

Most useful if you only have ten minutes: questions 1, 4, 8, 10 and 12. Question 4 changes how I state my main claim, 8 and 10 determine whether one of my core experiments is necessary at all, and 12 could change the framing of the paper.